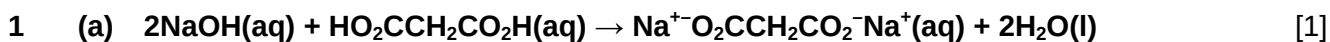


H2 Chemistry 9647**2013 AJC2 Prelim P2 Suggested Solutions**

correct state symbol given

(b) Intended [FA 2] [6]

Given that [malonic acid] = 1.00 mol dm^{-3} , and the proportion of solutions are 20 cm^3 **FA 1** and 40 cm^3 aqueous NaOH,

minimum [NaOH] in **FA 2** to ensure complete reaction of malonic acid
= 1 mol dm^{-3}

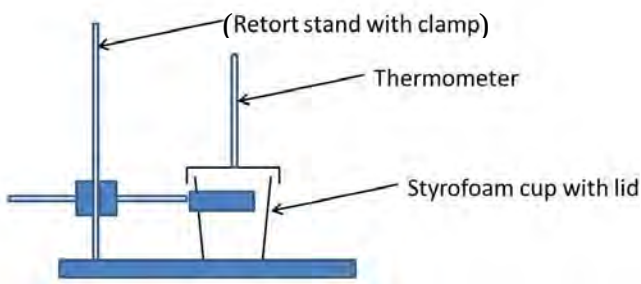
$n(\text{NaOH})$ required to prepare 250 cm^3 of **FA 2** = $1 \times 0.25 = 0.250 \text{ mol}$
mass of NaOH required = $0.25 \times (23.0 + 16.0 + 1.0) = 10.0 \text{ g}$

Therefore, about 11 g of solid anhydrous NaOH (more than 10 g) is used in the preparation of standard solution.

Preparation of FA 2

1. Weigh accurately about 11.0 g of solid anhydrous NaOH in a *pre-weighed weighing bottle*. Dissolve this solid in a (100 cm^3) beaker with 30 cm^3 of water.
2. *Reweight* the emptied weighing bottle.
3. Transfer the solution and washings into a *250 cm^3 graduated flask* and make up to the mark *dropwise* with deionised water.
4. *Shake the solution* to obtain a homogeneous solution and label it **FA 2**.

Experimental Set-up



Experimental procedure

5. Transfer 20.0 cm^3 of **FA 1** using a *burette* into a *styrofoam cup*. Place a thermometer in the solution and record the *initial temperature of FA 1*.
6. Using a *$50 / 100 \text{ cm}^3$ measuring cylinder*, measure out 40 cm^3 of **FA 2**. Place a thermometer in the solution and record the *initial temperature of FA 2*.
7. Quickly transfer the 40 cm^3 of **FA 2** into the styrofoam cup containing **FA 1**. Cover the cup with a lid and observe the maximum temperature rise on the thermometer.
8. Record the *maximum temperature reached*.
9. Repeat experiment until temperature rise is consistent to 0.2°C .

- [1] Appropriate concentration of FA 2 determined with calculations shown.
NaOH solution should be more concentrated than **FA 1** (1 mol dm^{-3}) to ensure complete neutralisation of malonic acid.
- [1] Use of weighing balance without TARE function (OR reweighing done)
Determine the mass of anhydrous NaOH that was successfully transferred from the weighing bottle to prepare the standard solution.
- [1] Use of volumetric flask to prepare standard solution of **FA 2**
Appropriate mass of NaOH used for 100 cm^3 or 250 cm^3 volumetric flask.
- [1] Correct drawing of set-up with appropriate choice of apparatus for the experiment for insulation.
Styrofoam cup with lid and thermometer, (retort stand)
- [1] Show measurement of volumes of solutions and initial temperatures of both solutions before mixing.
- [1] Correct procedures in determining the maximum temperature reached.

(c) (i) Weighted average of initial temperature = $\frac{(20 \times a) + (40 \times b)}{(20 + 40)}$

$$\text{Temperature change} = c - \frac{(20 \times a) + (40 \times b)}{(20 + 40)}$$

$$\text{Heat change} = 60 \times 4.2 \times \left[c - \frac{(20 \times a) + (40 \times b)}{60} \right] \quad [1]$$

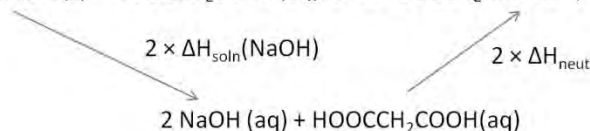
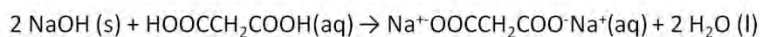
$$\text{Amount of water formed} = 2(0.02 \times 1) = 0.04 \text{ mol}$$

Enthalpy change of neutralisation

$$= -60 \times 4.2 \times \left[c - \frac{(20 \times a) + (40 \times b)}{60} \right] / 0.04 \quad [1]$$

$$= -105[60c - (20a + 40b)] \text{ J mol}^{-1}$$

(ii) ΔH_r [2]



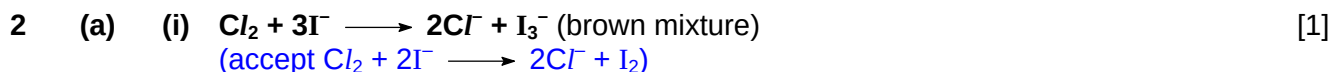
$$\begin{aligned} \Delta H_r &= 2 \times \Delta H_{\text{soln}}(\text{NaOH}) + 2 \times \Delta H_{\text{neut}} \\ &= (2)(-44.51) + (2)(-0.105)[60c - (20a + 40b)] \\ &= -89.02 - 0.21[60c - (20a + 40b)] \end{aligned}$$

[1]: correct cycle / application of Hess Law

[1]: correct ΔH expression in terms of **a**, **b** and **c**

- (d) solid anhydrous sodium hydroxide is highly corrosive and gloves should be worn. [1]

(accept: malonic acid and sodium hydroxide solutions are corrosive)



(ii) The organic layer is purple. [1]

Iodine formed, being more soluble in organic solvent, will dissolve in the trichloroethane to give a purple layer. [1]

(b) (i) 2 ppm = 2 mg of HOCl in 1000 g of solvent
 = 2 mg of HOCl in 1000 cm^3 of solvent

$$= \frac{2 \times 10^{-3}}{(1.0 + 16.0 + 35.5)} \text{ mol of HOCl in } 1 \text{ dm}^3 \text{ of solvent}$$

$[\text{HOCl}] = 3.8 \times 10^{-5} \text{ mol dm}^{-3}$ [1]

(ii) $K_a = \frac{[\text{H}_3\text{O}^+][\text{ClO}^-]}{[\text{HOCl}]}$ [1]

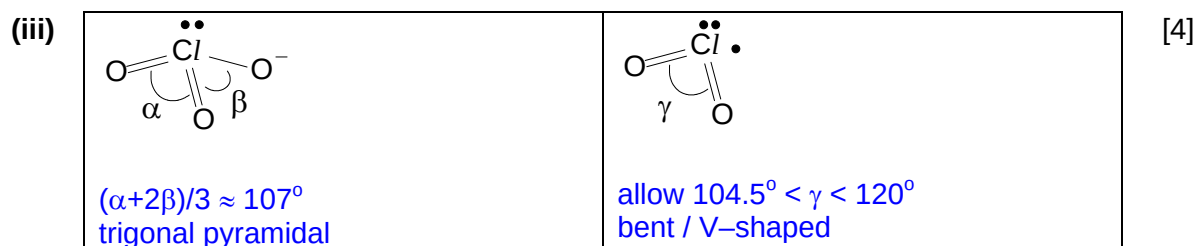
(iii)
$$[\text{ClO}^-] = \frac{K_a[\text{HOCl}]}{[\text{H}_3\text{O}^+]}$$

$$= \frac{(3.5 \times 10^{-8})(3.8 \times 10^{-5})}{(10^{-7.5})}$$

$$= 4.21 \times 10^{-5} \text{ mol dm}^{-3}$$
 [1]

(c) (i) O.N. of chlorine in NaClO_3 +5 [2]
 O.N. of chlorine in ClO_2 +4

[1] each



[1] each: correct shape

[1] each: correct bond angle



(ii) No. of moles of $\text{ClO}_2 = \frac{150}{[35.5 + 2(16.0)]} = 2.22 \text{ mol}$

No. of moles of $\text{Cl}_2 = \frac{1}{2}(2.22) = 1.11 \text{ mol}$

Mass of $\text{Cl}_2 = 1.11 \times [2(35.5)]$
 $= 78.8 \text{ g}$ [1]

3 (a) % of Hf by mass = $\frac{178}{178 + 2(16.0)} \times 100\% = 84.8\%$ [1]

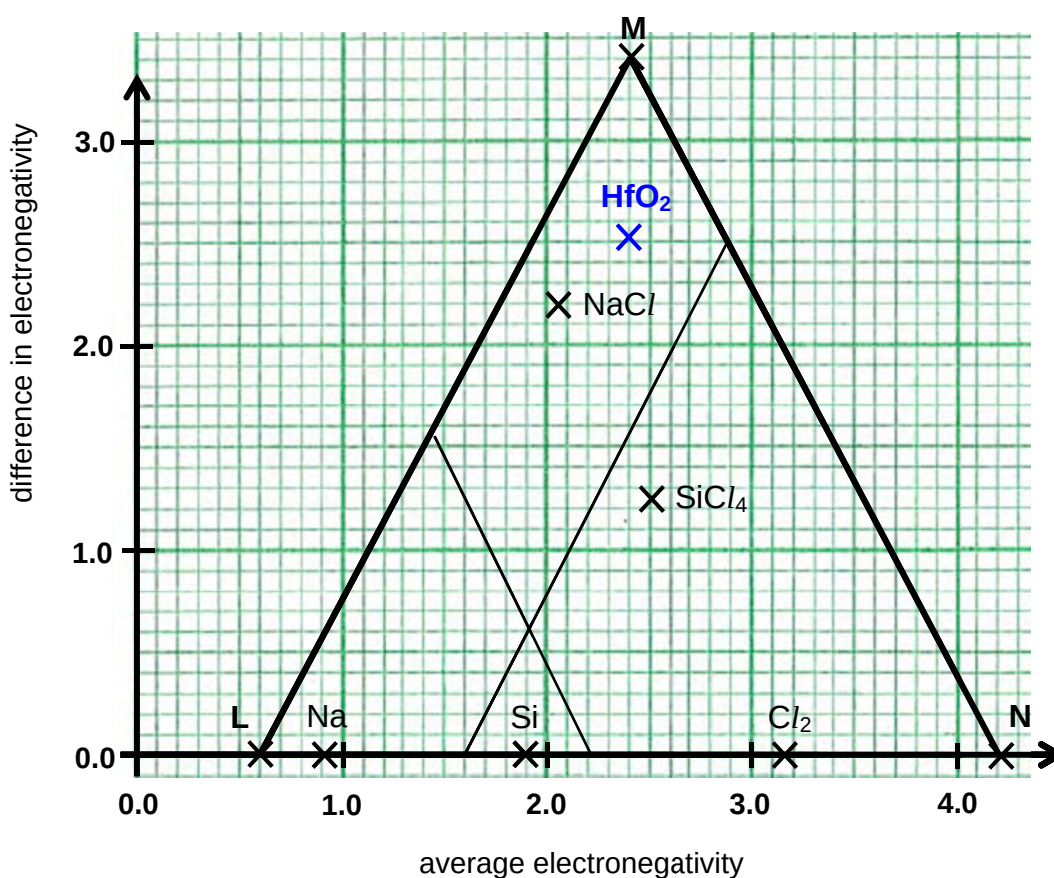
- (b) (i) L metallic [1]
M ionic
N covalent

(ii) average electronegativity for Hf and O = $\frac{1.16 + 3.61}{2} = 2.39$ (2.35 – 2.40)

difference in electronegativity for Hf and O = $3.61 - 1.16 = 2.45$ (2.45 – 2.50)

[1] each: correctly determined and plotted the coordinates for HfO_2

[2]

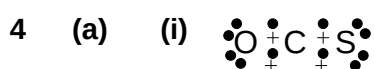


The oxide of hafnium is best described as **ionic** (circled)

[1]

(c) The oxide of hafnium is an **insulator** (circled)

[1]



[1]

(recall 'substitution' approach, OCS is similar to OCO)

(ii) 180°

[1]

(b)

temperature of the gas / °C	change in volume of the gas / cm ³	temperature of the gas / K	volume of the gas / cm ³
-23	-4.20	250	21.8
11	-2.10	284	23.9
29	0.40	302	26.4
63	3.30	336	29.3
94	7.00	367	33.0
131	9.20	404	35.2

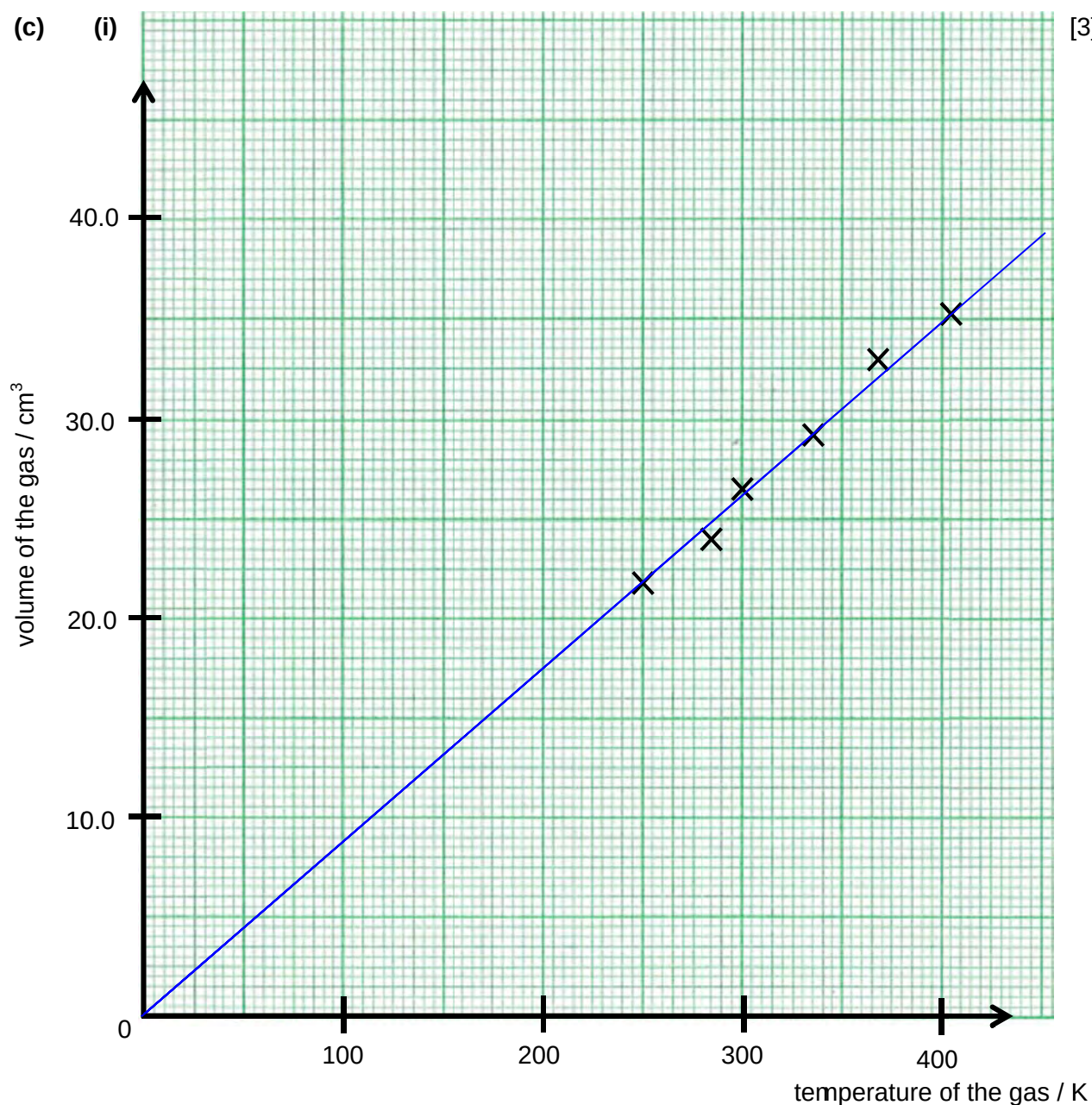
[2]

[1]: correct determination of readings

[1]: all readings to 3 s.f. (ignore error in rounding off)

(c)

(i)



[3]

[1]: label the x-axis temperature and the y-axis volume (with correct units). The plotted points must cover at least half the grid in both directions and all points must be on the given grid.

The point given by question (298, 26.0) can be included / excluded.

If a true origin has been scaled in and the student's line is extrapolated back to there then the origin is to be included as a "plotted point" in the assessment of covering half the grid in both directions.

[1]: check the plotting of points 1, 3, 6

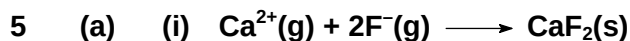
[1]: a 'straight-line of best fit' drawn (allow 'ecf' here from incorrectly plotted points).

- (ii) Charles's law is confirmed because a straight line passing through the origin (direct proportionality) is obtained. [1]

(The data confirms the relationship $V = kT$ or that volume is directly proportional to absolute temperature.)

Alternative

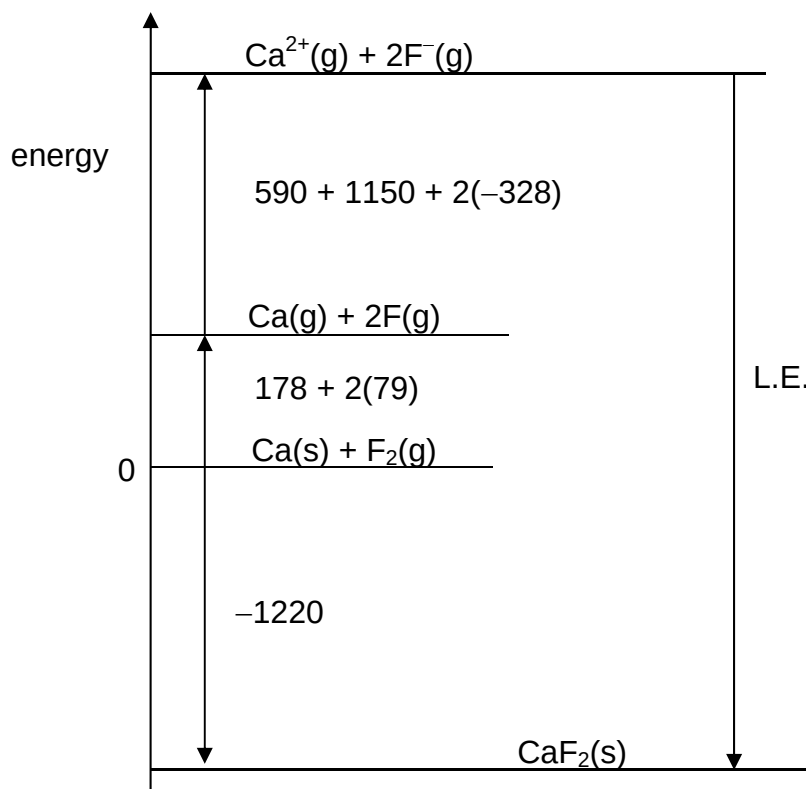
Calculations showing that V/T is the same for more than 1 point on the graph



[1]

(ii)

[3]



$$178 + 2(79) + 590 + 1150 + 2(-328) + \text{L.E.} = -1220$$

$$\text{L.E. of CaF}_2 = -1220 - 178 - 2(79) - 590 - 1150 + 2(328)$$

$$= -2640 \text{ kJ mol}^{-1}$$

[1]: correctly labelled cycle (in terms of ΔH or its value)

[1]: correct application of Hess' Law

[1]: L.E. of CaF_2 correctly determined

- (iii) Lattice energy of CaF_2 is less exothermic than that of FeF_2 as the ionic radius of Fe^{2+} (0.076nm) is smaller than that of Ca^{2+} (0.099nm). [1]

- (b) (i) Let $x \text{ mol dm}^{-3}$ be the solubility of CaF_2 in water.

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{F}^{-}]^2$$

$$3.2 \times 10^{-11} = (x)(2x)^2$$

$$x = 2.00 \times 10^{-4} \text{ mol dm}^{-3}$$

[1]

- (ii) $\text{pH} = 12 \Rightarrow \text{pOH} = 2$

$$\Rightarrow [\text{OH}^{-}] = 10^{-2} = 1.00 \times 10^{-2} \text{ mol dm}^{-3}$$

$$K_{\text{sp}} \text{ of } \text{Ca}(\text{OH})_2 = [\text{Ca}^{2+}]_{\text{total}}[\text{OH}^{-}]^2$$

$$6.4 \times 10^{-6} = [\text{Ca}^{2+}]_{\text{total}}(1.00 \times 10^{-2})^2$$

$$[\text{Ca}^{2+}]_{\text{total}} = 0.0640 \text{ mol dm}^{-3}$$

[1]

$$[\text{Ca}^{2+}]_{\text{total}}[\text{F}^{-}]^2 = 3.2 \times 10^{-11}$$

$$(0.0640)[\text{F}^{-}]^2 = 3.2 \times 10^{-11}$$

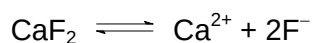
$$[\text{F}^{-}] = 2.24 \times 10^{-5} \text{ mol dm}^{-3}$$

$$\text{Solubility of } \text{CaF}_2 = \frac{1}{2}(2.24 \times 10^{-5})$$

$$= 1.12 \times 10^{-5} \text{ mol dm}^{-3}$$

[1]

- (iii) Solubility of CaF_2 in solution **Q** is lower than in water due to common ion effect. [1]



Presence of Ca^{2+} from $\text{Ca}(\text{OH})_2$ increases the $[\text{Ca}^{2+}]$, which shifts the position of the above equilibrium to the left, resulting in a decrease in the solubility of CaF_2 . [1]

(c) (i) $\Delta G_{\text{ppt}}^\ominus = 2.303 \times 8.31 \times 298 \times \log(3.2 \times 10^{-11})$
 $= -59.9 \text{ kJ mol}^{-1}$ [1]

(ii) $\Delta G_{\text{ppt}}^\ominus = \Delta H_{\text{ppt}}^\ominus - T\Delta S_{\text{ppt}}^\ominus$
 $\Delta S_{\text{ppt}}^\ominus = \frac{\Delta H_{\text{ppt}}^\ominus - \Delta G_{\text{ppt}}^\ominus}{-T}$
 $= -\left(\frac{-224 - (-59.9)}{298}\right) \times 1000$ [1]
 $= -551 \text{ J mol}^{-1} \text{ K}^{-1}$ [1]

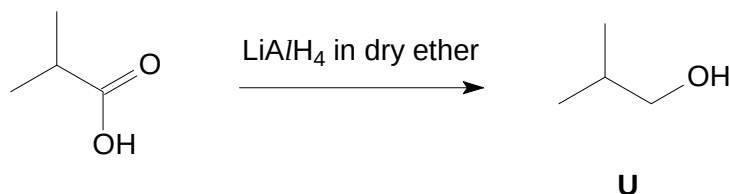
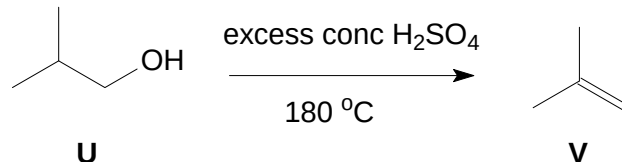
- (iii) $\Delta S_{\text{ppt}}^\ominus$ is negative when precipitation occurs.
 This is due to decrease in entropy during the phase change as the aqueous ions become more ordered in the solid precipitate. [1]

(d) $\Delta G_{\text{ppt}}^\ominus = 2.303(8.31)(298) \times \log(1.006)$
 $= +14.8 \text{ J mol}^{-1}$ [1]

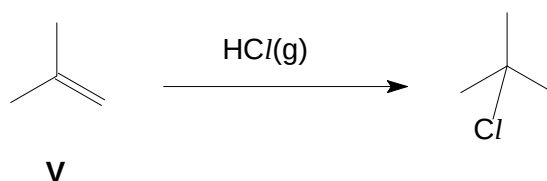
Since the $\Delta G_{\text{ppt}}^\ominus$ is positive, silver fluoride is soluble in water. [1]

6 (a) Step 1:

[5]

Step 2:

(accept other suitable dehydrating agent)

Step 3:[1] each: correct intermediates **U** and **V**

[1] each: correct reagents and conditions for each step

- (b) (i) Comparing experiments 1 & 2 in which $[\text{CH}_3\text{O}^-]$ is kept constant, when $[(\text{CH}_3)_3\text{CCl}]$ decreases by 5/6 times, the relative initial rate also decreases by $0.833/1.00 \approx$ 5/6 times. Hence, order of reaction with respect to $(\text{CH}_3)_3\text{CCl}$ is 1. [1]

Let the order of reaction with respect to CH_3O^- be x .

Comparing experiments 2 & 3,

$$\frac{(0.060)^x (0.050)}{(0.050)^x (0.040)} = \frac{0.833}{0.667}$$

$$\left(\frac{6}{5}\right)^x = 0.999 \approx 1$$

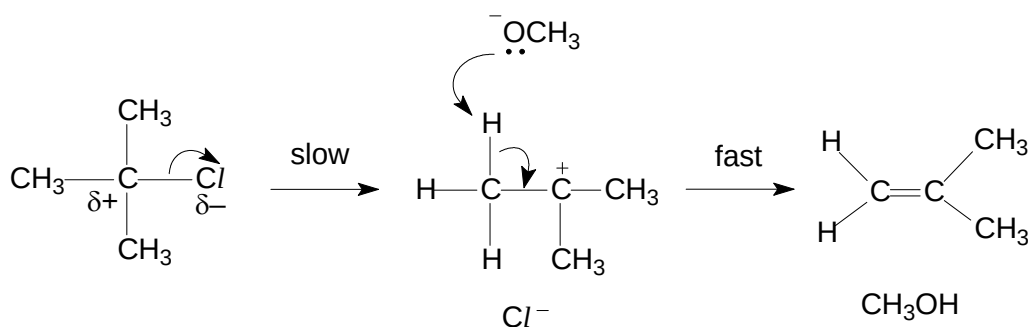
$$x = 0$$

[1]

Hence, order of reaction with respect to CH_3O^- is 0.Therefore, the proposed rate law is **correct**.

(ii)

[2]



[1]: correct electron pairs movement for both slow and fast step

[1]: correct intermediate and product generated

- (c) (i) rate constant will increase; [1]

at 100°C, the reactant molecules have higher kinetic energy and more molecules will have energy greater or equal to activation energy, resulting in an increase in rate of reaction and hence rate constant. [1]

(Since [reactants] remain constant, the increase in rate must be due to increase in rate constant.)

- (ii) rate constant will increase; [1]

since the rate-determining step involves the breaking of C–X bond, the use of *t*-butyl bromide will result in a lowered activation energy as C–Br bond is weaker than C–Cl bond. Hence the rate of reaction and rate constant increases. [1]

(accept use of Arrhenius equation $k = Ae^{-\frac{E_a}{RT}}$ in explanation for both parts)

- (d) X is (CH₃)₃COCH₃ [1]

- (e) (i) To prevent the amine formed (stronger nucleophile than NH₃) from undergoing further substitution with the RX. [1]

- (ii) [1]

